

Global Data Science Jobs Salary Analytics Dashboard

Interactive Business Intelligence Dashboard using Microsoft Power BI

Prepared By

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Project Type

Power BI Data Analytics Dashboard

Project Domain

Data Analytics & Business Intelligence

Tools Used

Microsoft Power BI, Microsoft Excel, Power Query, DAX, GitHub, Microsoft Word

Dataset Source

Kaggle - Global Data Science Jobs Salary Dataset

Dataset Format

Microsoft Excel (.xlsx)

Date

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1. Project Overview

The **Global Data Science Jobs Salary Analytics Dashboard** is an interactive Business Intelligence project developed using Microsoft Power BI. The primary purpose of this dashboard is to transform raw salary data into meaningful visual insights that help users understand salary trends across the global data science industry.

The dashboard provides valuable information about employee salaries, experience levels, company sizes, employment types, remote work trends, and job titles. Instead of reading thousands of rows of raw data, users can quickly explore information through interactive charts, KPI cards, and slicers.

This project demonstrates practical Business Intelligence techniques, including data cleaning, transformation, visualization, dashboard design, and KPI reporting.

Purpose of the Project

The main purpose of this project is to convert raw salary records into an interactive dashboard that enables users to explore salary trends efficiently. The dashboard supports quick decision-making by presenting business insights in a simple, professional, and visually appealing format.

Project Details

Project Title	Global Data Science Jobs Salary Analytics Dashboard
Project Type	Interactive Business Intelligence Dashboard
Project Domain	Data Analytics & Business Intelligence
Development Tool	Microsoft Power BI
Dataset Source	Kaggle - Global Data Science Jobs Salary Dataset
Dataset Format	Microsoft Excel (.xlsx)
Project Duration	Self-Learning Portfolio Project

Project Description

The dashboard allows users to explore salary information across different countries, job titles, experience levels, company sizes, employment types, work years, and remote work ratios. It uses KPI cards, charts, treemaps, donut charts, line charts, bar charts, and interactive slicers to provide a user-friendly analytical experience. The main objective of this project is to help users understand salary patterns, workforce distribution, and employment trends through professional data visualization techniques.

2. Problem Statement

Organizations collect large volumes of employee salary data every year. However, raw datasets are difficult to understand without proper visualization. Human Resource departments, recruiters, business analysts, and job seekers often need answers to questions such as:

- Which job title has the highest average salary?
- Which experience level earns the highest salary?
- Which company size hires the most employees?
- How has salary changed over different years?
- How common is remote work in data science?
- Which employment type is most popular?

Answering these questions manually requires significant time and effort. Therefore, an interactive Power BI dashboard was developed to transform complex salary data into clear, meaningful, and actionable business insights.

Organizations often collect large amounts of salary data, but raw datasets are difficult to understand. This project solves that problem by converting complex salary records into an interactive dashboard that helps users quickly identify trends and make data-driven decisions.

3. Project Objectives

The objectives of this project are:

- Analyze global data science salary trends.
- Compare salaries across different job titles.
- Identify salary differences based on experience levels.
- Analyze employee distribution across company sizes.
- Study employment type distribution.
- Understand remote work adoption.
- Monitor salary trends across different years.
- Design an interactive dashboard using Power BI.
- Build professional KPI cards.
- Enable users to perform self-service data analysis using slicers.

4. Project Scope

The dashboard analyzes global data science salary information based on the following dimensions:

- Countries
- Experience Levels
- Job Titles
- Company Size
- Employment Type
- Remote Work Ratio
- Work Year
- Salary

Users can filter the dashboard dynamically using slicers to analyze different scenarios.

5. Dataset Information

The dataset used in this project is the Global Data Science Jobs Salary Dataset, which contains salary-related information for professionals working in various Data Science roles across different countries.

The dataset was collected from Kaggle and imported into Microsoft Power BI using Microsoft Excel (.xlsx) format.

The dataset includes important attributes such as:

- Work Year
- Experience Level
- Employment Type
- Job Title
- Salary
- Salary Currency
- Salary in USD
- Employee Residence
- Remote Work Ratio
- Company Location
- Company Size

Before building the dashboard, the dataset was carefully reviewed to ensure consistency and usability for reporting and visualization.

6. Data Cleaning & Transformation

Data preparation was completed using Microsoft Power BI Power Query Editor. The following steps were performed during data cleaning and transformation:

- Imported the dataset from Microsoft Excel.
- Checked the dataset for missing or null values.
- Verified data types for all columns.
- Removed unnecessary formatting inconsistencies.
- Renamed columns where required for better readability.
- Loaded the cleaned dataset into the Power BI data model.
- Created DAX measures for KPI cards and dashboard calculations.
- Validated the dataset before designing the dashboard.

These steps ensured that the dashboard produced accurate and reliable business insights.

7. Dashboard Design Process

The dashboard was designed with a focus on clarity, usability, and professional visual presentation. The layout follows a logical flow: high-level KPI summaries at the top, followed by comparative charts, trend visualizations, and interactive filtering controls, allowing both quick-glance insights and deep-dive analysis.

8. KPI Cards

KPI cards were used to highlight key summary metrics at a glance, such as average salary, total number of employees/records, highest paying job title, and most common experience level. These cards give users an immediate snapshot of the overall dataset before exploring detailed charts.

9. Charts & Visualizations

The dashboard uses a combination of visuals to represent different aspects of the data:

- Bar charts - comparing average salary by job title and country.
- Line charts - showing salary trends across different work years.
- Donut/Pie charts - illustrating employment type and remote work distribution.
- Treemaps - visualizing company size or job title distribution.

10. Interactive Slicers

Slicers were added for key dimensions such as work year, experience level, employment type, and company size. These allow users to filter the entire dashboard dynamically and explore specific scenarios without needing to modify the underlying data.

11. Target Users

This dashboard is useful for:

- HR Professionals
- Recruiters
- Business Analysts
- Data Analysts
- Job Seekers
- Students
- Researchers

12. Business Value

By converting raw salary data into an interactive, visual format, this dashboard reduces the time required to identify workforce and compensation trends. It supports faster, more informed decision-making for HR planning, compensation benchmarking, and recruitment strategy.

13. Expected Outcome

After using this dashboard, users can:

- Understand global salary trends.
- Compare salaries across different job positions.
- Analyze workforce distribution.
- Explore interactive reports using slicers.
- Support business decisions with data-driven insights.
- Improve reporting efficiency through Business Intelligence techniques.

14. Skills Demonstrated

- Data cleaning and transformation using Power Query.
- Data modeling and DAX measure creation.
- Dashboard design and layout best practices.
- Business Intelligence reporting and KPI design.
- Data storytelling through interactive visualization.

15. Challenges & Solutions

During the development of this dashboard, several practical challenges were encountered.

Challenges

- Selecting the most suitable visualizations for different types of data.
- Creating accurate DAX measures for KPI cards.
- Designing a clean and professional dashboard layout.
- Arranging visuals within limited dashboard space.
- Configuring interactive slicers to filter all visuals correctly.

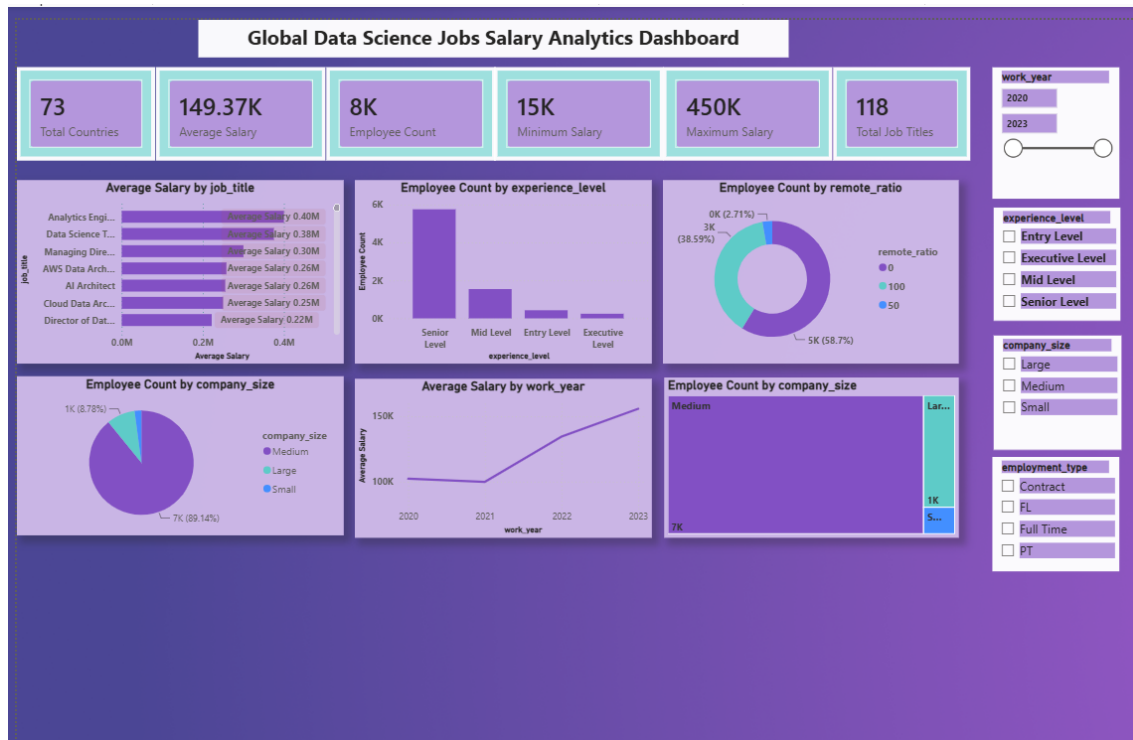
Solutions

- Appropriate Power BI visualizations were selected based on business requirements.
- DAX measures were tested and validated before implementation.
- A professional dashboard theme and consistent formatting were applied.
- Visuals were carefully organized to improve readability.
- Interactive slicers were tested to ensure correct filtering across all charts and KPI cards.

These improvements enhanced both the functionality and usability of the final dashboard.

16. Dashboard Preview

The following screenshot presents the final interactive Power BI dashboard developed during this project.



The dashboard provides interactive analysis through KPI cards, charts, and slicers, allowing users to explore salary trends across different dimensions.

17. Conclusion

The Global Data Science Jobs Salary Analytics Dashboard successfully transforms raw, complex salary data into a clear, interactive, and visually engaging Business Intelligence solution. Through the use of KPI cards, charts, and slicers, the dashboard enables HR professionals, recruiters, analysts, and job seekers to explore global data science compensation trends efficiently and make informed, data-driven decisions.

18. Future Improvements

- Integrate real-time or more frequently updated salary datasets.
- Add predictive analytics to forecast future salary trends.
- Include additional dimensions such as education level or specific technical skills.
- Publish the dashboard to Power BI Service for online sharing and collaboration.
- Add drill-through pages for deeper, role-specific analysis.

19. References

- Microsoft Power BI Documentation
- Microsoft Learn
- Kaggle - Global Data Science Jobs Salary Dataset
- Power BI Community
- Microsoft Excel Documentation